Linear Functions & Inequalities

ANSWER KEY

Section 1: Slope, Intercepts & Rate of Change

1 A line in the xy-plane passes through the points $(-3, 4)$ and $(5, -8)$. What is the slope of the line?

a. $-\frac{3}{2}$ ✓

b. $-\frac{2}{3}$

c. $\frac{3}{2}$

d. -1

Use the slope formula: $m = \frac{-8-4}{5-(-3)} = \frac{-12}{8} = -\frac{3}{2}$.

2 The linear function f satisfies $f(0) = 8$ and $f(4) = 20$. What is the value of $f(7)$?

a. 26

b. 29 ✓

c. 32

d. 35

The y-intercept is 8. The slope is $m = \frac{20-8}{4-0} = 3$. Therefore, $f(x) = 3x + 8$. For $x = 7$: $f(7) = 3(7) + 8 = 29$.

3 What is the x-intercept of the line given by $4x - 7y = 28$?

a. $(7, 0)$ ✓

b. $(4, 0)$

c. $(-7, 0)$

d. $(0, -4)$

Set $y = 0$: $4x - 7(0) = 28 \Rightarrow 4x = 28 \Rightarrow x = 7$. The x-intercept is $(7, 0)$.

Section 2: Parallel and Perpendicular Lines

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4 Line L_1 has the equation $3x + 2y = 10$. Line L_2 is perpendicular to line L_1 and passes through the point $(6, -1)$. What is the equation of line L_2?

a. $y = \frac{2}{3}x - 5$ ✓

b. $y = -\frac{2}{3}x + 3$

c. $y = \frac{3}{2}x - 10$

d. $y = -\frac{3}{2}x + 8$

Solve for y in L_1: $2y = -3x + 10 \Rightarrow y = -\frac{3}{2}x + 5$. The perpendicular slope is the negative reciprocal: $m_\perp = \frac{2}{3}$. Using point-slope: $y - (-1) = \frac{2}{3}(x - 6) \Rightarrow y + 1 = \frac{2}{3}x - 4 \Rightarrow y = \frac{2}{3}x - 5$.

5 Line P has equation $y = 4x - 9$. Line Q is parallel to line P and passes through $(2, 3)$. What is the y-intercept of line Q?

a. -5 ✓

b. -3

c. 5

d. 11

Parallel lines share the same slope $m = 4$. Using point $(2, 3)$: $3 = 4(2) + b \Rightarrow 3 = 8 + b \Rightarrow b = -5$.

6 In the xy-plane, line k passes through $(1, 2)$ and $(4, 8)$. If line m is perpendicular to line k, what is the slope of line m?

a. $-\frac{1}{2}$ ✓

b. $\frac{1}{2}$

c. -2

d. 2

Slope of k: $m_k = \frac{8-2}{4-1} = \frac{6}{3} = 2$. The perpendicular slope is $m_m = -\frac{1}{2}$.

Section 3: Single-Variable and Multi-Step Inequalities

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7 What is the greatest integer x that satisfies the inequality $5 - 3x \geq -16$?

a. 6

b. 7 ✓

c. 8

d. 9

Subtract 5: $-3x \geq -21$. Divide by -3 and reverse inequality: $x \leq 7$. The greatest integer is 7.

8 Solve the compound inequality: $-7 \leq 2x - 3 < 11$.

a. $-2 \leq x < 7$ ✓

b. $-5 \leq x < 8$

c. $-2 \leq x \leq 7$

d. $-4 \leq x < 7$

Add 3 to all parts: $-4 \leq 2x < 14$. Divide by 2: $-2 \leq x < 7$.

9 A cell phone company charges a monthly base fee of 30 dollars plus 0.08 dollars per megabyte of data used over the plan limit. If a subscriber wants their total bill to be at most 54 dollars, what is the maximum number of additional megabytes they can use?

a. 250

b. 300 ✓

c. 350

d. 400

Set up the inequality: $30 + 0.08m \leq 54 \Rightarrow 0.08m \leq 24 \Rightarrow m \leq \frac{24}{0.08} = 300$ megabytes.

Section 4: Two-Variable Inequalities & Systems

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10 Which of the following points lies in the solution set of the system $y > 2x - 1$ and $y \leq -x + 6$?

a. $(1, 3)$ ✓

b. $(3, 6)$

c. $(0, -2)$

d. $(4, 1)$

Test $(1, 3)$: $3 > 2(1) - 1 \Rightarrow 3 > 1$ (True). And $3 \leq -1 + 6 \Rightarrow 3 \leq 5$ (True). Both conditions hold.

11 The shaded region of an inequality in the xy-plane lies below the solid boundary line $3x - 4y = 12$. Which inequality represents this region?

a. $y \geq \frac{3}{4}x - 3$

b. $y \leq \frac{3}{4}x - 3$ ✓

c. $y \leq -\frac{3}{4}x + 3$

d. $y \geq -\frac{3}{4}x + 3$

Solve for y: $-4y = -3x + 12 \Rightarrow y = \frac{3}{4}x - 3$. Region below a line means $y \leq \frac{3}{4}x - 3$.

12 A farmer has at most 120 meters of fencing to enclose a rectangular vegetable garden. If the length must be at least 40 meters, what is the maximum possible width of the garden?

a. 15 meters

b. 20 **meters** ✓

c. 25 meters

d. 30 meters

Perimeter: $2L + 2W \leq 120 \Rightarrow L + W \leq 60$. Since $L \geq 40$, $W \leq 60 - L \leq 60 - 40 = 20$ meters.

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